Representational Design for Clinical AI: Embedding Secondary Stakeholder Needs in Large Language Models

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Abstract

Large Language Models (LLMs) are increasingly used to generate clinical documentation, yet most systems focus on easing individual users' clerical burdens rather than supporting multi-stakeholder work. We explore how LLMs expand the design space of clinical multi-stakeholder systems. Through a research-through-design study with 10 dermatologists using a teledermatology prototype, we explore how designing for secondary stakeholders opens new design opportunities and reshapes primary users' work. Applying the distributed cognition lens to our findings, we conceptualised representational design, i.e., the process of encoding secondary stakeholder perspectives (representations) in LLMs through, among others, prompts, templates, and constraints, while attending to LLMs' latent knowledge. We envision representational design as supplemental to data- and interaction-level approaches to designing multi-stakeholder artificial intelligence (AI) systems. We demonstrate how LLM-mediated representations cause misalignments between generated outputs and primary users' notions of acceptability, and how attending to these misalignments sets requirements for human-AI interaction techniques.

CCS Concepts

• **Human-centered computing** → **Collaborative and social computing theory, concepts and paradigms**; **HCI theory, concepts and models**; • **Computing methodologies** → **Artificial intelligence**.

Keywords

human-AI collaboration, multi-stakeholder AI, HCI, CSCW, HCAI, healthcare, reporting, clinical documentation, workflow, ML, machine learning, generative AI, genAI

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1 Introduction

Large Language Models (LLMs) introduce fundamentally new interaction possibilities for AI systems by expanding the design space through which AI behaviour can be shaped, interpreted, and negotiated. This has consequences for the domains it is applied to. In healthcare, LLMs are transforming clinical documentation work from a largely solitary writing task into a multi-perspectival communication challenge. Clinicians must now navigate an expanded set of editorial considerations generating patient-facing, lay-friendly explanations [59] and ensuring factual accuracy to support clinical continuity [22]. This shift is visible in emerging systems ranging from patient-facing conversational agents [52] to automated radiology reporting tools [53]. As such systems become more capable, it becomes increasingly important to attend to where and how their outputs circulate within clinical work. Designing clinical AI should therefore be understood as shaping elements of coherent healthcare pathways, rather than deploying isolated tools detached from practice [31, 64, 67, 68]. In this context, clinical reporting shifts from an individual burden of documentation to a design problem of representing, coordinating, and reconciling diverse stakeholder perspectives.

This work suggests that LLMs expand the design space of interactive systems by enabling new forms of human–AI interaction and introducing additional sites where design decisions can be made. Beyond interfaces and workflows, designers can influence how AI systems articulate, prioritise, and stabilise different perspectives within the artefacts they generate. In healthcare, understanding these new design possibilities becomes critical, as AI-generated documentation mediates care across organisational and professional boundaries. This motivates our research questions: *How do LLMs expand the design space for clinical multi-stakeholder systems?* and *What are the consequences of a multi-stakeholder approach to the primary users' work?* We define primary users, as people directly interacting with AI to produce certain output.

To investigate this question, we studied store-and-forward dermatology consultations. In a typical consultation, a primary care provider (PCP) submits a referral with clinical images and patient history. The dermatologist reviews this asynchronously, produces a written report, and sends it back. There is no real-time channel for clarification: if the report is ambiguous, incomplete, or pitched at the wrong level of detail for the receiving clinician, the PCP must either act on insufficient information or initiate a new referral cycle, making documentation a central site of multi-stakeholder coordination.

We conducted a research-through-design study with ten dermatologists using a teledermatology prototype that structured data entry and constrained LLM-generated reports through five information themes derived from a prior co-design study with PCPs [68]. In practice, dermatologists reviewed patient data, entered assessments, and iteratively edited LLM-generated drafts until reports met their standards. By embedding perspectives of secondary stakeholders (people affected by AI's output but not directly interacting with AI to produce it) directly into report generation, we could examine how new representational sites, rather than interface interactions alone, reshape primary users' work.

Drawing on Distributed Cognition (DC) [25, 28], we analysed the observed interactions as *coordination* across four conceptualised report representations: (1) the dermatologist's internal understanding of what the report should convey, (2) the LLM's latent model of dermatology and report conventions, (3) the PCP-derived template embedded to reflect downstream information needs, and (4) the externalised report draft produced by the system. While *coordination* in Computer Supported Cooperative Work (CSCW) often denotes the orchestration of tasks, resources, and people [37, 47], here we emphasise *coordination* as the process by which actors manage and align these heterogeneous representations to produce a satisfactory outcome. In practice, this took the form of dermatologists iteratively modifying both their input and the system's drafts until they were satisfied with the report. We extend DC's analytical foundation by showing how upstream *representational design* choices, specifically, embedding secondary stakeholder needs via templates, offer new opportunities for designers to shape the user experience before any interaction design. These representations reshape how primary users must align internal understandings, model behaviour, and externalised artefacts, thereby expanding the design space available to designers of multi-stakeholder LLM-based systems.

Based on this study, we make three contributions to HCI research on designing clinical multi-stakeholder LLM-based systems:

(1) We introduce *representational design* as a conceptual lens that complements data- and interaction-level approaches to multi-stakeholder AI design, showing how template-based operationalisation of secondary stakeholders influences the needs for downstream human-AI interaction techniques.
(2) We conceptualise five alignment dimensions that characterise how AI-generated reports may diverge from clinicians' expectations, offering a vocabulary for diagnosing representational misalignments in clinical AI systems. The alignments span: factual, confidence, importance, completeness, and style dimensions.
(3) We identify four coordination techniques that dermatologists employ to align these misalignments: input-level and word-level revisions, pruning for relevance, and internal representation alignment. We highlight how representational design shifts the burden and form of human–AI coordination.

The paper is structured as follows. Section 2 reviews clinical AI design for multi-stakeholder contexts, AI for clinical reporting, and relevant concepts from distributed cognition. Section 3 describes our research-through-design methodology. We present findings in two parts: Section 4 conceptualises five alignment dimensions, and Section 5 examines four coordination techniques dermatologists employed to achieve acceptable alignment. In Section 6, we establish *representational design* as a design approach for multi-stakeholder AI, discuss coordination as a productive process, and propose strategies for minimising misalignments. Section 7 and 8 conclude with our study's limitations and implications of *representational design*.

2 Related work

2.1 From single-user to multi-stakeholder clinical AI

AI in healthcare has become a central concern within HCI, with studies examining a wide range of clinical contexts and highlighting both opportunities and challenges. A recurring design challenge concerns the uncertainty and interpretability of AI-generated outputs. Within HCI and computer science, eXplainable AI (XAI) has emerged as a prominent response, focusing on methods and presentation techniques that help health professionals interpret model behaviour [40].

However, XAI largely frames explainability as a problem for a *single user* interacting with an AI system. This single-user orientation is also evident in broader human–AI interaction research, where design guidelines are typically articulated for an abstract "user" [3]. More recently, discourses on AI in health have shifted toward treating AI systems as sociotechnical rather than purely technical artefacts, foregrounding human-centred perspectives over model-centric ones [5, 68]. This shift has important implications for design, particularly a growing concern for how AI systems integrate with, reshape, or disrupt clinical workflows and divisions of labour. Beede et al.'s seminal study [8], for example, shows how a deep learning tool introduced into retinal screening disrupted nurses' work, leading to workarounds and, in some cases, system abandonment when contextual constraints clashed with the tool's assumptions. Subsequent research similarly emphasises that AI success depends on alignment with the temporal, social, and organisational rhythms of clinical work, including collaborative practices [67], everyday routines [63], and the capacity for local adaptation through reconfiguration of roles and tasks [48, 69].

Attending to workflows necessarily expands the design focus beyond a singular user to the plurality of stakeholders affected by AI systems, including those who may never directly interact with the technology. Research has argued for conceptualising clinical AI as "multi-user systems that support patient–provider collaboration" [30], and for involving key stakeholders throughout design to surface diverse needs and scenarios [60]. Studies that follow AI systems from early design through deployment tend to engage a broader set of stakeholders, including other medical professionals and non-clinical staff such as administrators and technicians [68]. The SepsisWatch project, for instance, illustrates how value is produced across organisational levels rather than at the level of individual users alone [10, 48, 49]. Complementing this perspective, scholars argue that patients and other indirectly affected stakeholders warrant meaningful participation, emphasising ideals such as informedness, consent, and agency in AI-mediated care [1, 65]. Yet in many clinical systems, these stakeholder concerns are not addressed through direct interaction design, but are instead embedded into artefacts and processes that mediate care.

Beyond stakeholder involvement, important design choices arise around how humans and machines are combined in practice [27, 62]. Wilson et al. [62], for example, characterise clinical human–AI systems along a spectrum from single-human–single-agent configurations to multi-human–multi-agent arrangements. As systems move toward the latter, additional interactional challenges emerge, including informational overlap, asymmetries in access to information, and decisions about how different stakeholder perspectives are prioritised and stabilised within shared AI-generated outputs [34].

More broadly within HCI and CSCW, research has long demonstrated that health technologies must accommodate multiple users with divergent perspectives and goals [56, 67]. Studies of patient-clinician interaction reveal tensions when perspectives diverge, such as between parents and clinicians, and show that systems only succeed when concerns are rendered meaningful and actionable for all parties [4, 33]. Similarly, research on multidisciplinary dashboards further illustrates how shared artefacts can both support and complicate coordination in chronic care settings [11]. Together, this line of work suggests that multi-stakeholder health technologies depend not only on interaction design, but also on how digital systems structure and organise collaborative work.

This aligns with the CSCW research agenda, that have long focused on the use and design of shared workflows and artefacts that structure how work is coordinated across roles and contexts [46, 47]. One key concept for understanding such environments is that of boundary objects. Boundary objects are artefacts that are sufficiently flexible to accommodate local needs while maintaining a shared identity across contexts [54]. Building on this perspective, work on boundary negotiating artefacts shows how shared artefacts are shaped and reconfigured in use as stakeholders negotiate meaning and action around them [12]. For example, work on personal informatics foregrounds how shared healthcare data can be interpreted and used differently by patients and providers while still functioning as a coordination artefact [12]. From this perspective, digital artefacts do not simply transmit information, but participate in aligning, translating, and sometimes unsettling shared understandings in collaborative clinical work.

2.2 AI and LLMs for clinical reporting

Large Language Models are increasingly deployed to generate clinical documentation, from radiology reports [7, 53] to dermatology consultation notes [52]. However, evaluation frameworks focus predominantly on single-user metrics: factual accuracy benchmarked against ground truth [66], hallucination detection, or efficiency gains for individual clinicians. This framing overlooks how clinical documentation participates in multi-stakeholder coordination, where reports must serve not only their immediate authors but also downstream providers making treatment decisions.

AI and LLMs hold potential to reshape clinical reporting by reducing clerical burden, improving workflow efficiency, and enhancing the communicative value of clinical notes [23, 64]. Historically, systems like Automatic Speech Recognition (ASR) systems aimed to enable physicians to dictate directly into electronic health records rather than relying on secretarial transcription. While these systems promised efficiency, in practice, they embedded clinicians in new forms of editing work, where errors must be corrected and text verified before approval. Such experiences show that automation in reporting may replace the effort of writing with an equal burden of oversight [24].

With LLMs, new possibilities have emerged alongside challenges. LLM-based systems provide support for summarisation, structuring, and editing documentation within clinical workflows [23], as well as improving patient-clinician communication [34, 72]. However, based on evaluations of ambient AI scribes [50], the concerns about trust and authority tint the enthusiasm for reduced burden. Moreover, AI scribes and report generators increasingly position clinicians as editors of machine-generated drafts rather than sole authors. As argued by Altschuler et al. [2], this reframes reporting as an editorial task, carrying new responsibilities for narrative accuracy and clinical accountability.

Recent work examining LLM-based clinical documentation begins addressing these coordination dimensions. Studies explore how discharge letters function as coordination artefacts linking physicians, patients, and external providers [71], how communication challenges span professional boundaries in radiology and dermatology [14], and how automation can be reframed to serve report recipients rather than focusing solely on author efficiency [70]. However, this work does not examine the consequences of incorporating secondary stakeholder needs through AI. It establishes that LLM-based clinical reporting reshapes authorship, introduces new forms of editing work, and increasingly addresses multi-stakeholder contexts. As a result, the coordination work that emerges when AI-generated reports must serve asynchronous clinical workflows across professional boundaries remains insufficiently understood.

2.3 Distributed cognition as lens for human-AI interaction

Distributed Cognition theory in HCI views cognition as distributed across individuals, artefacts, and the physical and social environments rather than residing solely in individual minds [25, 28]. Cognitive processes emerge from interactions among people, tools, and culturally embedded practices, captured by DC's three tenets: cognition is (1) socially distributed, (2) embodied, and (3) culturally shaped. This positions DC as a framework for analysing collaborative work through the coordination of actors, representational structures, and artefacts. DC treats these elements as functionally integrated parts of a system organised around accomplishing a specific task. This perspective enables analysis of how knowledge and decision-making are distributed across and shaped by the interactions within the system [25].

In clinical work, DC reveals how knowledge and decision making propagate across representations, such as clinical documentation, images, forms, and conversations, each encoding information in ways that enable or constrain downstream action [41]. This perspective is particularly useful for LLMs, which embed and transform diverse representations, thereby challenging traditional boundaries between tools and agents in collaborative work.

Recent work in human-AI collaboration shows that effective collaboration requires aligning AI outputs with users' mental models and task contexts, and that AI systems' representation of unobservable information can shape how humans integrate model outputs into decision making and reasoning (e.g., [26]). However, existing research typically focuses on individual interactions and does not address the coordination challenges arising when AI systems embed the needs of absent (secondary) stakeholders through templates or generation constraints, a significant gap given the unique coordination demands of multi-stakeholder AI systems.

Our study addresses this gap by using DC to characterise coordination work in an LLM-based teledermatology system where PCP information needs were operationalised through template-constrained generation. We examine how dermatologists (primary users) coordinated across different report representations present in the cognitive system to produce reports satisfying both their professional standards and secondary stakeholder requirements.

3 Method

We conducted a research-through-design study with an LLM-based store-and-forward dermatology prototype to examine how LLMs expand the design space for clinical multi-stakeholder systems and what the consequences of a multi-stakeholder approach are to the primary end users' work.

3.1 Operationalising secondary stakeholder needs

The prototype was designed to embed PCP information requirements identified in prior research into the reporting activities of dermatologists [70]. That study used co-design methods with PCPs to identify five information themes (the PCP template) required in teledermatology reports to support patient care: *Visual features*, *Diagnosis*, *Treatment plan*, *Alternative plan*, and *Follow-up*.

We operationalised these downstream information needs through two interconnected design mechanisms:

Structured input form: Dermatologists entered clinical assessments via a diagnostic input form with fields corresponding to the five PCP information themes (Figure 1). This design made secondary stakeholder priorities visible at the data-entry stage, foregrounding downstream information needs where they are typically absent in specialist reporting workflows.

Template-guided generation: The LLM was prompted to generate reports following a structured template matching the PCP information themes. The prompt included: (1) the five-section template structure, (2) definitions of each section taken from [70], (3) the dermatologist's input mapped to corresponding sections, and (4) optional examples demonstrating expected content (Figure 2). This ensured all generated reports included information prioritised by PCPs (if clinically relevant), regardless of dermatologists' input style or completeness.

Through these mechanisms, the prototype embedded secondary stakeholder needs into both the input interface and the AI generation process, creating a system where PCP information requirements shaped dermatologists' work from data entry through final report approval.

3.2 Prototype design

The prototype simulated a store-and-forward teledermatology workflow through three interface components: a patient data viewer, a structured diagnostic form, and an editable report area (Figure 1).

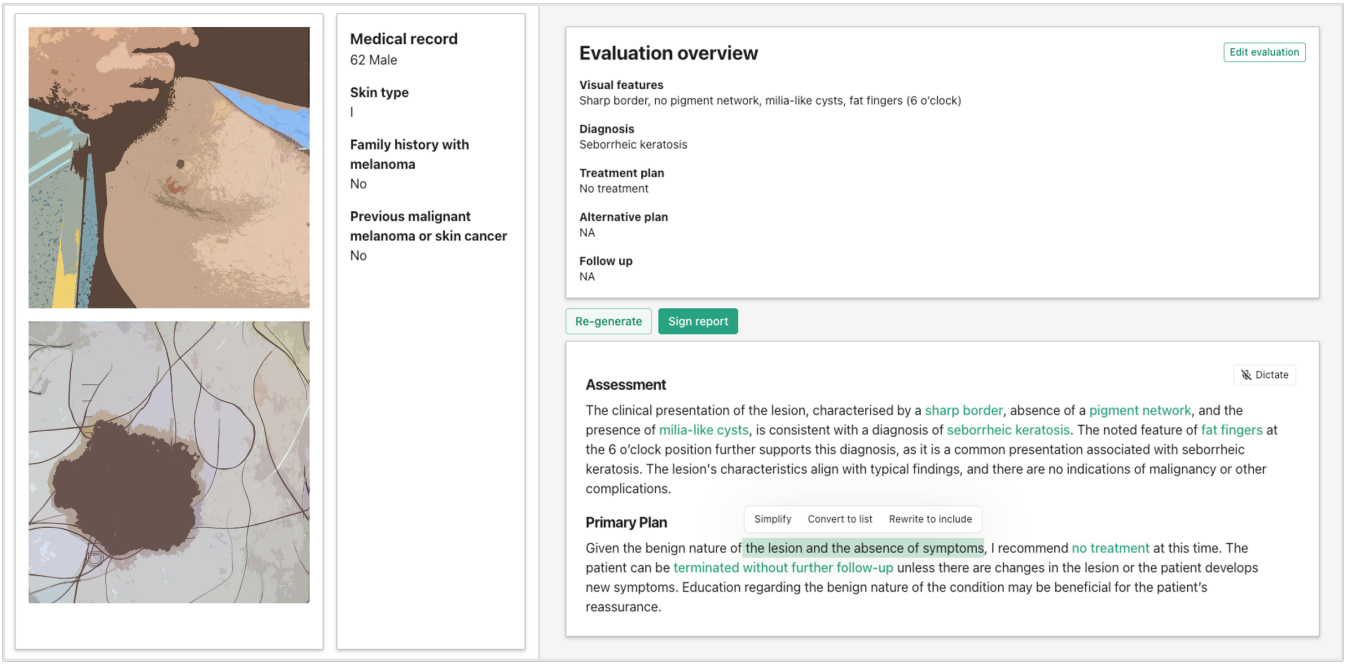


Figure 1: Prototype design. Screen with obscured Case 1 (patient images and clinical history). On the right-hand side, there is an approved diagnostic form (rendered as read-only *Evaluation overview*). Below, there is a generated report with a highlighted phrase and visible contextual interactions.

The patient data viewer displayed anonymised clinical and dermoscopic images with zoom functionality, along with patient clinical information. This component replicated the typical information dermatologists receive in store-and-forward consultations.

The diagnostic form collected dermatologists' clinical assessment via fields corresponding to the five PCP information themes. All fields were optional and free-text, allowing dermatologists to express their clinical reasoning flexibly. To focus the study on report generation rather than diagnostic accuracy, fields were prefilled with assessment data prepared by NKT (see Section 3.3). Participants could modify or replace this content as needed. The completed form served as input for LLM report generation.

Report area displayed LLM-generated reports structured according to the PCP template, with each section populated from corresponding diagnostic form entries. Reports could be edited via free-text modification or built-in functions that varied across iterations (see below): early iterations offered "simplify," "convert to list," and "rewrite to include" functionalities executed by the LLM, whereas later iterations added contextual menus, system prompt exposure, and optional few-shot example toggles. Dermatologists retained full editorial control: they could modify, delete, or rewrite any content, reflecting their professional and legal responsibility for clinical documentation. Although the system generated reports following the five-section PCP template, dermatologists were not prevented from removing sections or restructuring the format. However, the diagnostic form's structure, the LLM's consistent template-based generation, and the report area's visual organisation all operationalised PCP information themes as the default and expected structure. These made deviation from this format an active choice rather than a neutral option.

Iterative refinement of human-AI interactions. While the patient viewer and diagnostic form remained constant across all sessions, we iteratively refined the report area to better support dermatologists' coordination work. We updated the prototype after every two sessions until Session 7, where no major usability issues were reported (four iterations total, covering all ten participants). These iterations focused on improving human-AI interaction mechanisms while maintaining the core design constraint: all reports consistently addressed the five PCP information themes.

- **Iteration 1 (Participants 1–2):** The report area displayed generated text divided into five paragraphs matching the PCP template sections. Each paragraph could be swapped for pre-generated shorter or longer alternatives. Free-text editing was supported.
- **Iteration 2 (Participants 3–4):** Instead of presenting as two separate paragraphs, which caused repetition, "Visual description" and "Diagnosis" were rendered as a single "Assessment" paragraph. Critically, both information themes remained present. This was a presentational change, not a content reduction. The diagnostic form still collected input for all five themes, and all five were addressed in the generated reports. Introduced few-shot prompting to supplement theme definitions in the generation prompt. Added paragraph-scoped "summarise" and "convert to list" functions.

```
const systemPrompt = `
You are an experienced dermatologist preparing a
    professional medical report for a general
    practitioner referral.
Context and expectations:
   Use precise, concise medical language
   Demonstrate clear clinical reasoning
   Ensure all information is clinically relevant
   Write in a professional tone, using appropriate
       medical terminology
   Write in first person
   Don't include skin type if not relevant`

const casePrompt = `## Report structure
   The report must include the following sections: ${
       getSections(sectionDefinitions)}
   ${generatePatientPrompt(patient)}
   ## Condition details: ${evaluationPrompt}
   ## Formatting requirements
   - Use <h2> tags for section headings and <p> tags
       for paragraphs.
   - Highlight input data used in the report with <
       strong> tags.
   ${generateExamplesPrompt(sections, flags.
       includeExamples)}`

const sectionDefinitions = [{
   "title": "Assessment",
   "description": "Your interpretation of the visual
       findings in the light of the referral and
       your problem understanding"
   },...]
```

Figure 2: A list of prompts specifying LLM's behaviour. First, a system prompt. Below, a dynamic prompt structure utilising PCP template (*sectionDefinitions*). generatePatientPrompt(*patient*) includes patient data. *evaluationPrompt* contains the dermatologist's diagnostic form. generateExamplesPrompt is a method that conditionally includes examples prepared by NKT for few-shot prompting. *sectionDefinitions* is a JSON object containing the PCP template definitions. All definitions adopted from [70], here an example of the Assessment (/emphvisual findings + diagnosis input fields).

Iteration 3 (Participants 5–6): Removed pre-generated paragraph alternatives. Made regeneration functions contextual: they affected selected text rather than entire paragraphs. Introduced "rewrite to include" function, allowing dermatologists to specify content to incorporate.

Iteration 4 (Participants 7–10): Exposed the system prompt, allowing dermatologists to view and modify LLM generation instructions. Made few-shot examples optional toggles rather than defaults, giving dermatologists control over generation style guidance.

These refinements explored different approaches to supporting coordination work while preserving the fundamental design constraint: the PCP template remained unchanged across all iterations. All generated reports continued to address downstream PCP information needs regardless of which interaction mechanisms dermatologists employed.

The prototype was implemented as a React.js web application, with GPT-4o and 4o-mini handling report generation via OpenAI's API.

3.3 Study design

The study simulated real-world store-and-forward teledermatology workflows (Figure 3). Each participant reviewed six anonymised cases obtained from Melatech: three skin growths and three rashes. Each case included anonymised clinical and dermoscopic images (skin growths only) with synthetic patient histories composed by NKT. To focus observation on report generation coordination rather than diagnostic work, all diagnostic form fields were prefilled with clinical assessments prepared by NKT. This was a deliberate design choice to reduce domain uncertainty and isolate coordination demands produced by the embedded representations rather than by diagnostic disagreement. We accepted a loss of ecological validity because the prototype necessarily simplified teledermatology work: participants worked with limited clinical information and without routine interruptions, time pressure, or diagnostic uncertainty. To reduce friction cause by predefined diangoses, participants could modify or replace this content as needed, and were instructed to do so if they disagreed with the assessment. NKT selected cases representing varying complexity (simple, complex, and poor image quality) for each condition to reflect real-world practice diversity. This number and variety provided rich observational data while keeping sessions to approximately one hour. This approach allowed us to isolate coordination work arising from the PCP template structure without confounding it with diagnostic variability.

Before beginning, participants received an introduction to the study's purpose, completed a background questionnaire, signed informed consent, and were familiarised with the prototype's functionalities and the embedded PCP template structure.

For each case, participants:

(1) Reviewed patient data as in their daily practice, examining images and clinical information through the patient data viewer.
(2) Entered clinical assessments via the structured diagnostic form, which corresponded to the five PCP information themes.
(3) Generated initial reports by triggering LLM generation based on their (modified or original) diagnostic form input. The system produced reports structured according to the PCP template.
(4) Reviewed and edited reports using free-text modification or available AI interactions. Participants decided when reports satisfied their clinical standards by accepting the report. This judgment was left to their professional discretion and constituted a key aspect of our observational focus. We did not specify acceptance criteria, as understanding what dermatologists considered "satisfactory" when balancing their own

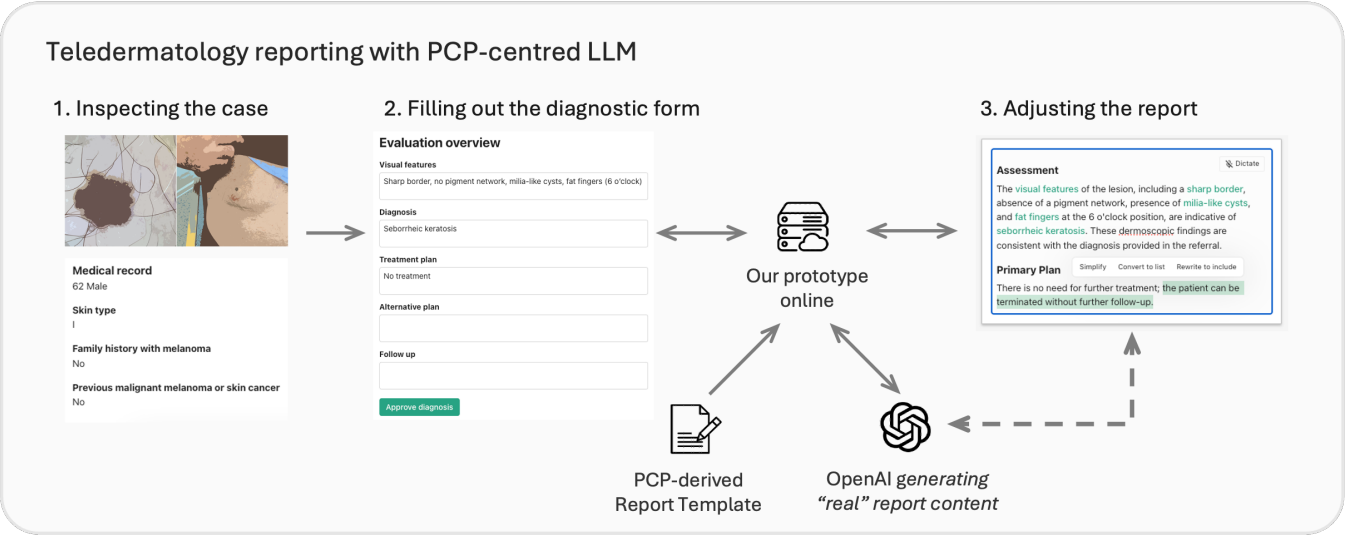


Figure 3: A flowchart of the sessions with our participants. Each participant completed six realistic cases within three steps: inspecting the case, approving / filling out the diagnostic form, and editing the generated report until approval. Participants could repeat the steps if necessary. The flowchart specifies the use of LLM within the session.

standards with embedded PCP requirements was central to our research question.

Throughout this process, participants were instructed to think aloud, verbalising their reasoning as they navigated between their clinical understanding, the diagnostic form structure, LLM-generated outputs, and the evolving report drafts. This allowed us to observe the coordination work required to align these multiple representations.

Participants were not compensated. The study was classified as non-interventional interview research under Danish law and therefore did not require university or national research ethics committee approval.

3.4 Participants

We recruited 10 dermatologists from Denmark (4), Sweden (2), and the United States (4) using convenience sampling, based on availability and access through professional networks. All participants were medical doctors specialised in dermatology and had prior experience with teledermatology.

3.5 Data Collection

We collected data through think-aloud observation [39] and semi-structured interviews [32]. All sessions were online and lasted approximately 58 minutes on average. We collected screen recordings, including audio of all interactions with the prototype across 60 case reports (10 participants × 6 cases each). This included dermatologists' modifications to prefilled diagnostic forms, report generation triggers, editing actions (free-text modifications and function use), and final approval decisions. Due to the small participant pool, we prioritised qualitative depth over quantitative metrics, focusing on patterns of coordination work rather than statistical measures of interaction frequency. Think-aloud protocols captured participants' verbalisations as they worked through each case, revealing reasoning about clinical assessments, reactions to LLM-generated content, and decision-making processes as they navigated between the diagnostic form, LLM outputs, and evolving report drafts. Finally, semi-structured interviews explored participants' experiences with the PCP-focused system. Topics included: current teledermatology practices, perceived usability of the prototype and AI interaction techniques, quality and accuracy of generated text, the scope and completeness of reports, alignment with downstream PCP needs, and concerns about clinical responsibility and authorship. All sessions were video-recorded, AI-transcribed, and manually corrected for accuracy. The study took place between November 2024 and March 2025.

3.6 Data Analysis

We analysed data using an iterative approach combining descriptive coding with theory-driven interpretation through Distributed Cognition (see Section 2.3 for theoretical foundation). This integrated method enabled us to both characterise dermatologists' experiences and develop theoretical insights into how the embedding of secondary stakeholder needs reshaped their coordination work.

KB and CWN transcribed video recordings using AI-assisted transcription (Dovetail), manually corrected for accuracy, and independently coded all sessions. Following familiarisation with the material, we conducted descriptive coding capturing: dermatologists' interactions with the prototype, their strategies for aligning LLM outputs with clinical reasoning, reflections on the embedded PCP template, and notable observations from think-aloud protocols. This produced 347 codes, which were exported to Miro[1] along with associated quotes.

[1] miro.com

ID	Dermatology experience (years)	Tele-dermatology experience	Country of practice
P1	20 y	Ca. 1000 consults	Sweden
P2	15 y	Ca. 5000 consults	Sweden
P3	16 y	Ca. 5000 consults	USA
P4	16 y	Ca. 1000 consults	Denmark
P5	30 y	Ca. 1000 consults	Denmark
P6	23 y	Ca. 5000 consults	Denmark
P7	20 y	Ca. 5000 consults	Denmark
P8	4 y	< 100 consults	USA
P9	19 y	Ca. 500 consults	USA
P10	12 y	Ca. 5000 consults	USA

Table 1: Participant background, a consult – a completed consultation including reporting

KB and CWN conducted five rounds of thematic clustering between February and May 2025, supported by weekly discussions with HDZ. These meetings iteratively refined the conceptual structure through dialogue with the data and our emerging observations. During this process, we explored trust and efficiency in AI-assisted documentation, additional work, perceived report usefulness, tensions between clinical ownership and standardisation, and supporting PCP efficacy.

Following a two-month washout period to reduce initial interpretive framings' influence, HDZ revisited all codes with explicit DC focus. We applied DC's analytical constructs to our empirical observations: representational states, cognitive off-loading, and cultural influence. We mapped DC's core concepts onto our empirical observations: treating the PCP template, LLM outputs, diagnostic form, and report drafts as representational states across which dermatologists coordinated. We examined how cognition was distributed between dermatologists' clinical expertise and the AI system's capabilities. Through systematic comparison of observed behaviours against DC concepts, we identified theoretically productive tensions: instances where coordination work deviated from expected patterns or revealed novel mechanisms. For example, we observed dermatologists simultaneously editing for multiple audiences (PCPs and themselves), negotiating between LLM-generated phrasing and their own clinical voice, and developing workarounds when template structure conflicted with case-specific reasoning. We iteratively tested these emerging insights against the full dataset, refining our characterisation through two cycles of theoretical mapping.

4 Representations misalignments

A central observation when embedding the goals of PCPs into an LLM-based teledermatology reporting system was the ambivalent acceptance of generated reports. Opinions ranged from equivocal enthusiasm and a desire for immediate integration to dissatisfaction and outright rejection of such technology.

Distributed cognition provides a useful analytic tool for understanding the different outcomes, as it treats clinical reporting not just as an individual act of writing, but as a system of cognitive work in which information is transformed across multiple representational states. This theoretical framing is useful, as it introduces new representations that shape the structure and communication of clinical reports. These representations can be internal, such as a dermatologist's mental model of how a report should be structured, or external, such as the written report itself.

The process of writing clinical documentation with an LLM is therefore not a straightforward expression of the dermatologist's intent, but rather the result of an LLM's propabilistic transformation of dermatologist's input through the introduced representations. This transformation almost inevitably results in a misalignment between the dermatologist's internal representation and the external representation of the generated report (Fig. 4).

4.1 Different report representations in clinical reporting with LLMs

To understand how these misalignments arise, it is necessary to unpack the specific representations that LLMs introduce into the clinical work. In conventional reporting, dermatologists directly translate their internal understanding of a case into an external (composed) report, often with minimal structural and semantic constraints from the tools they use. In our study, however, this translation was mediated by additional representational layers: the PCP template and the LLM's latent representations of dermatology knowledge. In total, we were working with four representations of a single report within this functional system.

The most essential and obvious one was the reporting dermatologists' internal representation, i.e., a mental model of how a case should be described and what constitutes a good report. This representation guided reasoning, decision-making, and finally reporting, informed by clinical data, images, and PCP referrals. Participants based their report vision, among others, on training, guidelines, experience, and expectations of recipient needs. This internal representation formed the foundation of the reporting process, against which other representations, whether templated, modelled, or generated, were evaluated and coordinated.

Next, the LLM generating the report was always embedded with knowledge about dermatology from the training data. We will refer to it as LLM's latent representations. These representations surfaced when the model generated text. Despite deliberate prompting and temperature set to 0, the unpredictable nature of *every* LLM may cause hallucinations, data leakage, or unintended deviations ([51, 57]). As such, the LLM's latent representations (regardless of a specific model's performance) constitute a distinct layer in the

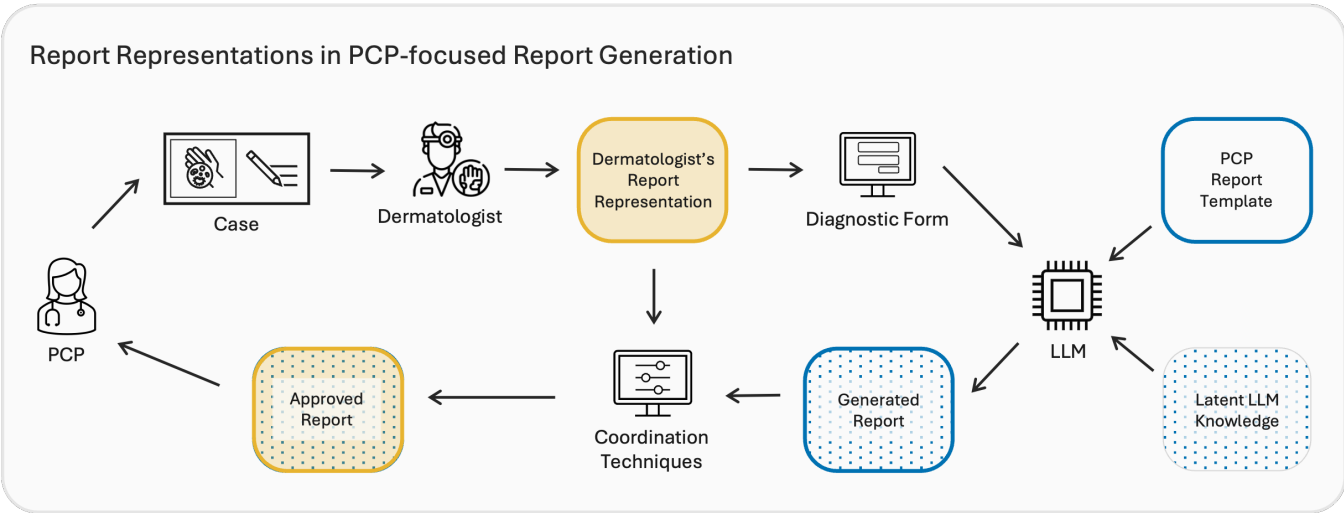


Figure 4: A schematic representation of a teledermatology workflow using LLM-based report generation with a PCP template. The schematic includes the system elements (icons) and the report representations conceptualised during the study (rounded rectangles).

reporting system, complementing but also potentially conflicting with dermatologists' internal models and PCP templates, thereby shaping the final report in unforeseen ways.

A further representation, unique to our setup, was the PCP template. This template reflected the expectations of primary care physicians regarding the content and structure of the teledermatology reports. By making the needs of report recipients explicit, the template had the potential to improve *informational continuity* [22] between clinicians, thereby strengthening patient care pathways. It also added a representational layer that may conflict with dermatologists' reporting practices, creating friction and extra coordination work.

Finally, the external representation was the report generated by the LLM, produced through the transformation of the dermatologist's input according to the prompt aligned with the dermatologist's intent, the dermatologist's use of human-AI interaction techniques, the LLM's latent representations embedded in its training data, and the PCP template reflecting recipient needs.

4.2 Misalignment of representations

When multiple representations of a single report coexist within a cognitive system, what patterns of misalignment emerge? An AI-generated report revealed the misalignments between the primary user's criteria for a good report and the representations introduced at this stage of work that influence the report's generation. We identified five such alignment dimensions (Tab. 2).

4.2.1 Factuality of the generated reports. This dimension was assessed against dermatologists' clinical knowledge, and any misalignments were the most serious errors, as even subtle factual inconsistencies when passed on to PCPs not only defeat the purpose of the consultation but also put the patient at risk. In this way, even small factual errors could acquire disproportionate significance, as they may travel downstream and influence current and future treatment decisions.

The most grave example of such an error observed was hallucinating a diagnosis. Despite the precautions taken, e.g., through prompting and providing all the required input, when reviewing the report for Case 5, P4 noted,

> *It... puts in its own diagnosis, 'irritant and allergic contact dermatitis', which is not something that the [diagnostic form] mentions at all.* [P4, Case 5]

This illustrates how the generated report emerges as a resultant representation, in which the primary user's perspective does not always take precedence.

Importantly, factual misalignments did not arise solely from LLM hallucinations. They often resulted from the interaction between dermatologists' inputs, the model's latent knowledge, and the template structures guiding report generation. Individual sentences or phrases in the reports could be correct in isolation, yet inconsistencies became apparent when considered together. For example, in Case 3 (P3, Figure 5), the system correctly recommended a shave/tangential biopsy for an exophytic tumour, but the margin assessment it added applies only to excisional biopsies. Here, the AI layered inappropriate details onto an otherwise correct procedure, creating a subtle contradiction that could misdirect clinical decision-making. Similarly, when the input from the diagnostic form was not sufficiently structured or explicit, the generated report could misrepresent it.

These examples illustrate how misalignments arise not from isolated errors but from the complex interplay between expert input, model latent knowledge, and external report templates, highlighting the hybrid nature of generated reports and the vigilance required in their use.

Type	Definition
Factuality	The report is factually correct
Certainty	The report expresses the intended level of certainty or uncertainty
Importance	The report conveys the appropriate clinical importance of the described conditions and observations
Completeness	The report includes the right amount of clinically relevant information
Style	The report reflects the intended tone, structure, and word choices

Table 2: Five dimensions of alignment between generated reports and dermatologists' internal report representations

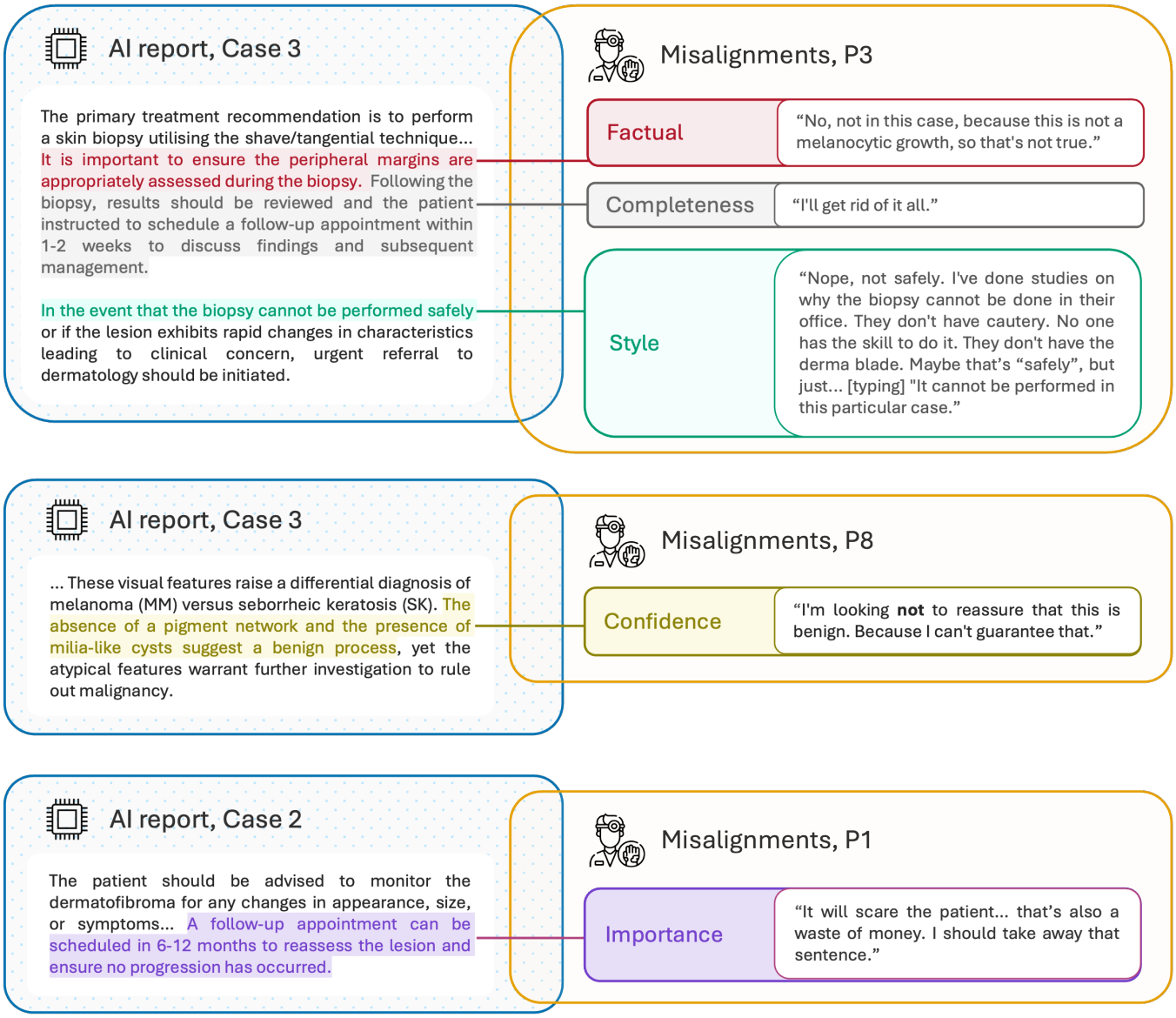


Figure 5: Examples of misalignment between AI-generated dermatology reports and dermatologists' visions for those reports. A single report can be misaligned across several dimensions.

4.2.2 Certainty expressed in the report's judgments. Communicating levels of *certainty* was seen as a critical aspect of clinical reporting, since recipients base treatment decisions not only on the clinical content but also on the dermatologist's confidence in their assessment. Both over- and under-confident framings carried risks with significant clinical and resource implications. For instance, an overconfident dismissal may cause patients to ignore future changes, while hedged phrasing prompts vigilance without undue alarm, noted by P9:

> *It says it's consistent with. It doesn't say it's definitely this... And then you have a hedge comment where, if it changes, it's symptomatic; they should come back in for an in-person examination because you can never be 100% sure on just a virtual examination.* [P9, Case 1]

Conversely, under-confident framings could trigger unnecessary follow-up appointments, increasing patient anxiety and straining healthcare resources, as observed in Case 3 (P8, Figure 5). These findings demonstrate that *certainty* is not merely a linguistic nuance but a crucial alignment dimension that shapes how clinical judgments are interpreted and acted upon throughout patient care pathways.

But how was certainty represented in our system? In our prototype, AI was not tasked with making predictions. The only way it could infer an appropriate level of *certainty* was through: prompting (explicit preference for more or less direct framing), its latent representations (correlation of conditions with typical ways of framing), and the diagnostic input indicating level of certainty. However, the diagnostic form contained only keywords participants considered relevant, rarely including explicit markers of confidence. The only indirect signal the model could rely on was the presence of multiple differential diagnoses, which made it difficult to align with the nuanced levels of certainty dermatologists held in their internal representations.

On the reader's side, expectations around *certainty* varied considerably, shaped by the potential consequences of a condition, the dermatologist's diagnostic confidence, and their comfort with directness. We also observed cultural and systemic influences: US-based participants often leaned towards hedging to manage liability and patient expectations. In contrast, European participants were more comfortable with concise and direct framings. These findings underscore that certainty is not simply a property to be extracted from inputs but a negotiated representation, distributed across clinicians, systems, and the contexts in which reports circulate.

4.2.3 Completeness of the information captured in the report. A clinical report should be *"short but complete"* [P10] – a seemingly straightforward but ultimately subjective requirement. Misalignments around reports' *completeness* were observed on a cursory–superfluous spectrum. More often, on the superfluous end. As P2 described, many reports contained *"too much yapping."* During iteration 1, this verbosity stemmed from the repetition of information across different sections included in the PCP template, which divided the report into discrete categories. While helpful in structuring the diagnostic form, these divisions did not align with dermatologists' practices of writing integrated, interlinked reports. This was later addressed by structuring the content of "Visual description" and "Assessment" themes in a single paragraph. The template inadvertently encouraged information to "leak" across sections, leading to redundancy and undermining one of LLM's potential benefits: generating concise yet rich narratives.

The model's cautious style further contributed to excessive length. Unlike our participants, who were accustomed to determining when advice can be definitive and when it must remain conditional, LLMs often defaulted to reiterating standards of care or recommending follow-ups. In Case 3 (P3, Figure 5), this produced a report that not only felt overly cautious to P3 but also risked sounding patronising and diminishing the PCP's agency. At the opposite end, some reports were too brief, usually because the diagnostic form provided limited information. This was highlighted by P10, who, after exploring the system prompt used in our prototype, explained that

> *The report is not intended to be a manuscript or a textbook or something like that, but sometimes it requires a little bit of explanation to ensure that they [PCPs] receive the information with the nuance that it requires. And so if the system is prompted to focus on brevity... it could be possible... it doesn't quite adequately convey the information that I want the primary care doctor to have.* [P10, Case 4]

Ultimately, these findings demonstrate how *completeness* is continuously negotiated between brevity, comprehensiveness and guidance, and shaped not only by individual judgment but also by external constraints, such as templates and input formats. By foregrounding PCPs' perspectives in the template, our study further shifted this balance, mediating how dermatologists' expertise was enacted. Achieving an appropriate balance remains a central challenge: too little information risks clinical gaps, whereas too much risks obscuring the most salient aspects.

4.2.4 Importance assigned to diagnoses described in the report. Assigning importance is part of how dermatologists communicate what deserves attention and what can be safely ignored. This judgment goes beyond naming conditions. It reflects an assessment of future risks, developments, and the relative weight of competing possibilities. In our study, this dimension often intersected with certainty: the more confident a dermatologist was, the more definite they were in dismissing benign findings and highlighting serious ones.

In some cases, however, the model misascribed importance. For example, in Case 2 (Fig. 5), it assigned greater importance to a benign condition and proposed additional follow-up. Such advice would only make sense if the dermatologist were uncertain about the diagnosis or if the lesion carried a plausible risk of malignancy – neither of which applied. The effect was not an outright factual error but a distortion of emphasis that risked burdening PCPs, patients, and the healthcare system with unnecessary monitoring.

Misascribing importance mattered because it shifted how information was acted upon downstream. Overstating the relevance of benign conditions could generate avoidable anxiety and follow-up work, while underplaying malignant risks could obscure what was most clinically urgent. From a distributed cognition perspective, importance served as a representational cue that coordinated attention and action across the care pathway. Even when the facts were

correct, getting the prioritisation wrong disrupted this coordination, reshaping how expertise flew between dermatologists, PCPs, and patients.

4.2.5 Style of the report. *Style* encompassed many aspects, including report structure, preferred wording, and overall tone. Most style-related misalignments were acceptable or could be overlooked unless they affected understandability or risked infringing on the professional relationship between dermatologists and PCPs. In Case 3 (P3, Figure 5), for example, a dermatologist flagged that describing a limitation to carrying out a biopsy as a matter of "safety" was misleading, as it softened what was in fact an infrastructural constraint. P3 seemed to want to ensure that the issue was not about trust in the PCP's competence, but rather whether they had the necessary resources and training to perform the procedure. This is why, in this particular case, they decided to edit this specific sentence.

Structural differences were also noted, but they were rarely seen as deal-breakers. For example, one report combined clinical and dermoscopic presentations into a single description, as noticed by one participant:

> *So we have the clinical and the dermoscopic presentation together... It's not false. And I would be happy to send it. It will work totally fine, but it's a little academically wrong, I guess. It's not 100% correct.* [P7, Case 2]

Although the structure diverged from P7's own practice, the absence of clinical consequences made it acceptable.

At the same time, many dermatologists reported that the generated reports often aligned well with their stylistic preferences. One dermatologist noted:

> *I think it said in words that I could be writing and I think it has captured what I wanted to explain... I think it's exactly what I would like to have written myself... I like that.* [P1, Case 5]

Strikingly, this alignment was achieved without providing writing samples, only through prompting. This suggests that when other dimensions such as factual correctness, certainty, and completeness are aligned, stylistic personalisation becomes less critical, with many dermatologists content as long as the report "sounds right."

In this sense, style appeared to be the most flexible of the dimensions: deviations were tolerated provided they did not compromise meaning, relationships, or care pathways.

5 Coordination of representations

Why did the use of LLMs in clinical reporting create a need for coordination? The main reason lay with responsibility. Regardless of technological support, dermatologists remained accountable for the reports they sign – legally, professionally, and personally. As P10 explained:

> *I feel like this is my responsibility... I'm putting my name on this. I own this. This is my decision-making. And ultimately, regardless of whether I type out every word or AI, or I'm copying and pasting it from a document that I have. I own the responsibility.* [P10, Case 3]

Distributed cognition helped to make sense of this tension by drawing attention to the role of cognitive artefacts. Cognitive artefacts, whether algorithms, calculators, or spreadsheets, are external objects that participate in cognitive processes by structuring, storing, or transforming information [25]. They may take over specific tasks, but their behaviour is stable and their outputs predictable. Their value lies in providing trustworthy, reliable, and accountable functionality [13]. LLMs do not follow the same set of rules. While they can be delegated the task of drafting a report, their probabilistic nature means their outputs cannot be anticipated with certainty. They can hallucinate, deviate from prompts, or introduce stylistic and factual choices that dermatologists neither intended nor endorsed. This complicates the notion of *cognitive offloading*. While offloading in distributed cognition does not necessarily reduce cognitive effort, especially in conjunction with LLM-based systems, it redistributes it. A task once handled entirely by the clinician may be partially transferred to an artefact, but new forms of coordination work arise in return: checking the accuracy of generated text, correcting misleading phrasing, and aligning the output with one's professional judgment.

As a result, responsibility remained firmly with the dermatologist, but control over the reporting process was only partial. This disjuncture created opportunities for misalignment between internal and external report representations, making coordination between them a central aspect of reporting with LLMs.

5.1 The goal of coordination

> *It captures what I was thinking.* [P6, Case 1]

The goal of coordination in our study was to align the dermatologist's internal representation of a report with its external, generated version so that it could be sent to the receiving PCP. Crucially, we found that *perfect alignment was not necessary*. Reports were often accepted as soon as they were factually sound and broadly reflected the dermatologist's intent, even if style and completeness diverged from their own preferences. In other words, while factuality was non-negotiable, other misalignments could be tolerated or adjusted with minor edits.

This flexible approach illustrated how responsibility was enacted in practice. Dermatologists did not aim to reproduce their internal representation verbatim but to ensure that the external representation could serve its purpose of supporting PCPs in patient care. As P3 explained, there are already tools in practice that mediate the individual style of doctors' writing. Thus, the focus should not be placed on replicating the fine details of doctors' unique communication patterns:

> *I don't think they're [PCPs] gonna believe that I wrote every word of it. Because what they are accepting of now is whether I see a patient [remotely] or I see a patient face to face. We all have smart phrases, right? No one expects me to write a separate care note every time. That's just nuts. But what they don't want is something that doesn't tell a story. They would say: "This tells the story of this patient and what I'm supposed to do..." It is clear. It is patient-centred, and it is concise enough that it's not annoying.* [P3, Case 1]

However, achieving acceptable alignment required meaningful work. For some participants, the coordination effort exceeded what they perceived as worthwhile. As P9 pointed out:

> *I don't find that it's particularly streamlined my work. No. I've had to type in the diagnosis and the treatment plan, and it spit out something. I still had to redo it. It's like double my work. I would have rather just dictated everything in the first place and be done.* [P9, Concluding remarks]

Yet when we adjust our baseline from the fastest possible way of composing a report to writing the best possible report for recipients, the cost-benefit balance shifts. P6 explicitly addressed this trade-off, emphasising the value of improved information continuity between dermatologists and PCPs:

> A1: *How do you compare this process to your current practice?*
> P6: *Much better... Because it would take me maybe five minutes to write that [more than with our prototype], and that report to the GP is much more useful... I think the feedback is that they [PCPs] would like to have more guidance... And the system appears to provide the GP with better guidance.* [P6, Concluding remarks, edited for clarity]

Coordination, then, became less about writing the report and more about managing the alignment of representations across multiple dimensions—balancing efficiency with quality, personal style with recipient needs, and speed with thoroughness. Such an alignment threshold was elusive, varying with the case, practitioners, and likely other factors. Nonetheless, it was always present, confirmed each time the dermatologist pressed the "Sign report" button. Successful coordination produced reports that were not only consistent with the dermatologist's internal standards but also aimed to support the information needs of PCPs.

5.2 The coordination work

Participants coordinated missalignments in four recurring ways: input-level revisions, word-level revisions, pruning for relevance, and adjustments of their internal representation (Fig. 6).

5.2.1 Input-level revisions. In cases where misalignments were caused by issues with the diagnostic input, such as ambiguity, omissions, or mistakes, dermatologists often revised the diagnostic form rather than rewrote the generated report. As P6 noted:

> *I think it's probably easier to edit the evaluation and then let the machine make a new report.* [P6, Case 3]

To illustrate it, in Case 3, P2 initially omitted a key visual feature. Rather than editing the generated report directly, they added it to the form and regenerated the whole report. They explained that they hoped the system would produce more aligned output without redoing all the prior work. By improving the clarity of the input data, participants were able to align the generated report with their vision without re-assuming the full cognitive task of report composition.

This behaviour illustrated how the division of cognitive labour influenced the coordination of representations within the system. Once dermatologists offloaded report composition to the LLM, rewriting entire paragraphs would require them to reassume work they had already delegated, imposing the extra cognitive cost of switching roles and incorporating their own writing into a coherent albeit unsatisfactory report. Instead, they acted on the information they embedded in the diagnostic form, refining and augmenting it to guide the LLM. This demonstrated how alignment can be achieved through incremental, input-level adjustments rather than complete task reassignment.

These findings suggested two key design considerations for LLM-based clinical tools. First, systems should support the division of cognitive labour, allowing dermatologists to focus on their expertise rather than forcing them to re-assume offloaded tasks. Second, LLMs for clinical documentation tools should rely on comprehensive input methods that enable the encoding of the richness of clinical knowledge. By concentrating on these aspects, designers could reduce wasted effort, minimise misalignments, and better align AI-generated outputs with clinicians' expectations.

5.2.2 Word-level revisions. Some of the misalignments identified by the participants were not due to issues with the diagnostic form data, but rather to the LLM's word choices, whether due to hallucinations or simply suboptimal phrasing. These errors could affect any of the five alignment dimensions and were typically confined to single words or short phrases that were not tightly connected to the surrounding text. For example, when reviewing a report P8 explained:

> *The word I'm kind of hung up on is 'the removal'. Because whenever we do this kind of biopsy, we don't guarantee the removal of the lesion.* [P8, Case 1]

In such cases, dermatologists preferred to edit the text directly rather than adjusting the form.

This differs from input-level revisions because it did not require re-assuming the full cognitive task of report composition. Instead, dermatologists were performing localised edits, adjusting wording or small phrases without restructuring the overall report or re-integrating the content into a broader cognitive framework. This means that the misalignment at the point of word-level revisions was usually less prominent than with input-level revisions.

For this scenario, we explored several human-AI interaction techniques, such as offering more concise or verbose alternatives, regenerating passages for brevity, inserting additional terms, or reformatting text into bullet points. Yet these techniques rarely resolved the misalignments. The changes required at this point were localised and lexically precise, making them ill-suited to additional rounds of opaque, unpredictable generation. In practice, dermatologists found that direct manual editing was faster, more predictable, and more precise. Additionally, even though they were not part of our study, many participants mentioned using other tools in their daily reporting, such as speech-to-text systems, macros, and templates, that could support such word-level revisions.

Building on these findings, we suggest that designers should treat word-level coordination as an editorial task rather than a generative one. Instead of layering more automation at this level, systems should make manual editing effortless through features such as inline text editing, transparent highlighting of generated versus user-edited text, and audit trails for accountability. Supporting direct editorial control recognises that, once content has been generated, expertise is best applied through fine-grained human judgment rather than further delegation to an AI model.

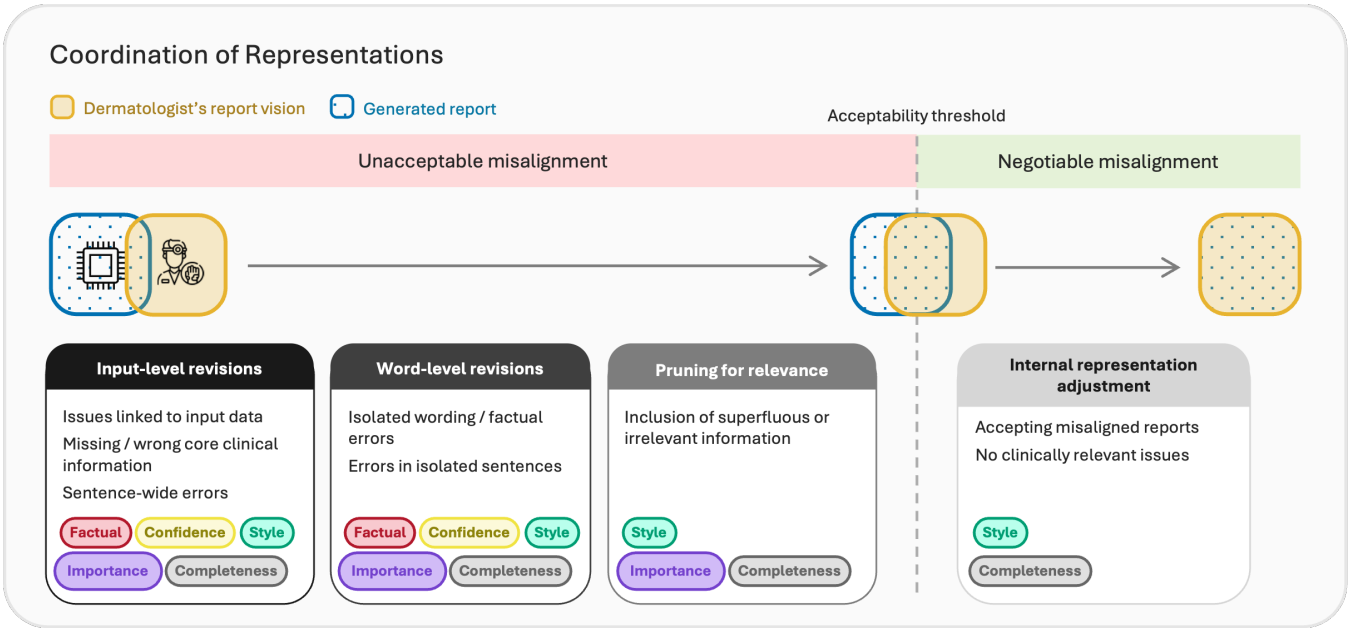


Figure 6: Flowchart of observed techniques for coordinating internal and external report representations, distinguishing unacceptable from negotiable misalignments, with more severe misalignments requiring more intensive coordination.

5.2.3 Pruning for relevance.

> *You don't want words in there that could be confusing or fluff that is going to take up someone's time.* [P3, Case 5]

A recurring observation among participants was that the reports were excessively long. Importantly, this length did not stem from dermatologists' extensive guidance, but from the way the LLM structured the text: wordy phrases, repetitions, and frequent safety disclaimers inflated the reports. In the first prototype iteration, we imposed no constraints on report length, and regardless of the input provided, each section of the generated report tended to be of similar size, suggesting that the model padded content to maintain structural balance. In these cases, participants often responded by simply deleting superfluous text.

We later introduced stricter system prompts to limit report length, thereby reducing some redundancy. Yet one challenge persisted: the model's inherent risk-aversion. Even with more direct prompts, reports often remained bloated by excessive safety recommendations. This pattern highlights that pruning for relevance is distinct from word-level revisions: the misalignment here lies not in individual word choices but in the systematic verbosity of the generated representation.

5.2.4 Internal representation adjustment. was the final coordination strategy we observed. It occurred when no clinically significant misalignments were present, and the remaining differences were related to style or completeness. These discrepancies posed no risk to patients but diverged from what dermatologists might have written themselves. Rather than making additional edits, participants often accepted the generated report as is, effectively aligning their internal expectations with the external representation. For example, P6 noted:

> *It didn't take in the fat fingers at 6 o'clock.* A1: *... would you add it?* P6: *I don't think so. I think I would just accept this... it looks nice.* [P6, Case 1]

This illustrates that perfect alignment across all dimensions was not necessary for report acceptance. In our study, the critical dimensions were factuality, confidence, and importance. Once these were secured, participants entered the space of acceptable misalignment (Fig. 6), where coordination shifted from modifying the external report to adjusting internal expectations, since further refinements in completeness or style were often not worth the additional effort. In other words, they determined which aspects of their internal representation they were willing to compromise in order to complete the report. This suggests that AI-based systems should not aim to enforce exhaustive alignment but instead support clinicians in recognising when "good enough" has been reached.

6 Discussion

6.1 Shaping human-AI interactions through representational design

Representational design reconfigures the conditions under which human-AI interaction becomes necessary. When designers encode secondary stakeholder perspectives, e.g., through prompts, templates, and constraints, they preconfigure the generative space and, in doing so, determine where and how misalignments will surface. These frictions are less about hallucination than about the commitments baked into representational choices, whose information needs are privileged and whose conventions are used. Consequently,

human-AI interaction in such systems is less about correcting errors than about coordinating heterogeneous representations toward an acceptable outcome. We argue that representational design in clinical AI systems thus precedes and shapes interaction design. It conditions misalignments, thereby setting the space for the required interaction techniques through which users reconcile multi-perspectival demands in AI-generated clinical documentation.

This work resonates with the CSCW concept of boundary objects. In our study, a generated clinical report can be read as a boundary object: it must carry meaning for dermatologists, PCPs, and patients, each interpreting it through different professional conventions. But boundary objects are descriptive. They explain how coordination emerges at the intersection of practice, not how designers can configure it in advance [54]. Representational design offers the prescriptive complement: designers decide upstream which stakeholder perspectives are formalised, how they are prioritised, and how they are operationalised in the generative system. Crucially, LLMs make this configuration qualitatively different from static template design. Templates deterministically constrain output through fixed fields, whereas LLMs probabilistically integrate representations, so perspectives can be combined and their surface expression emerges from generation rather than being enumerated by the designer. The novelty is therefore not that the report coordinates across different worlds, as boundary objects have always done, but that LLMs allow designers to preconfigure that coordination upstream, before any artefact exists.

Prior HCI work has addressed adjacent issues through interface design and participatory methods. Jacobs et al. argue that clinical decision support tools should accommodate multiple users simultaneously, but their focus remains on UI-level support rather than internal representational structure [31]. Delgado et al. similarly emphasise stakeholder involvement across the AI lifecycle, while noting that such participation often remains consultative and therefore has limited influence on how outputs are structured [15]. Ajmani et al. and Yoo et al. further highlight the importance of secondary and downstream stakeholders and the risks of neglecting them [1, 65]. Together, this work clarifies who should be considered in AI design. Representational design addresses how those perspectives can be encoded directly into the generative system itself.

Building on this work, our study suggests that coordination is not just a concern related to participation or interaction.

By designing representations, selecting which ones to incorporate into the system, and specifying how the AI should interpret and integrate them, designers can pre-shape the AI output and the subsequent interaction space. For example, in our case, the PCP template expanded the focus of the reports to benefit PCPs, while also requiring additional coordination work. The concepts of representations and representativeness are not new in machine learning or critical AI scholarship. For example, Bergman et al. [9] examine these notions in the context of datasets used for AI evaluation. They highlight that while careful attention to representation can support fairness and inclusivity, oversimplifying the concept risks objectification or exploitation. Moreover, achieving adequate representation in datasets often entails high costs and uncertain outcomes. In contrast to the costly efforts to capture stakeholder needs in datasets and model training, LLM-based systems make it relatively easy to introduce, update, or swap representations. This opens opportunities to embed stakeholder perspectives at workflow stages where they would otherwise be absent. This "middle way" can supplement data-level and interaction-level design. We demonstrate how upstream representations can extend the benefits of AI to secondary stakeholders, pre-shape AI outputs, and, in turn, the human-AI interactions and coordination practices that follow. We therefore suggest that representational design offers new opportunities for HCI researchers engaging with AI in multi-stakeholder settings, and we see three promising directions:

(1) **Developing methods for designing representations:** how can stakeholders' knowledge and needs be elicited and formalised for incorporation into LLMs?
(2) **Investigate embedding practices:** how should such representations be integrated into LLM-based systems to maximise the multi-stakeholder benefits while minimising primary users' need for coordination work?
(3) **Supporting transparency of influence:** How can systems communicate the relationship between specific elements of an AI-generated output and the representations that shaped them, supporting primary users' evaluation and coordination?

The framework developed here may extend beyond teledermatology to LLM-based system where outputs must serve multiple stakeholders with divergent information needs. For example, remaining in healthcare, AI scribes in primary care present a structurally analogous but reversed configuration: the GP is the primary user generating a referral, while the patient becomes the secondary stakeholder whose informational needs are encoded upstream [2, 50]. Radiology reporting involves a single generated document serving different clinicians like surgeons, oncologists, and referring physicians simultaneously. In this context factuality and completeness dimensions carry the highest consequence, given the safety-critical implications of omission or ambiguity [64]. Discharge summaries introduce a further complication, in that patients and GPs emerge as simultaneous secondary stakeholders with divergent needs, making style and certainty calibration the more consequential alignment dimensions [71]. We expect the five alignment dimensions identified in this study: factual, confidence, importance, completeness, and style, to transfer across these domains, though their relative severity and the coordination demands they impose on primary users would shift accordingly. Representational design is therefore not a fixed prescription but a context-sensitive practice, one whose application requires understanding who the secondary stakeholders are, what accountability structures govern primary users' work, and which misalignments are most consequential in each setting.

6.2 Coordination needs as a consequence of designing for secondary stakeholders

Initially, we anticipated that the challenge of integrating LLMs into clinical reporting would lie in designing the right editing interaction tools that would enable dermatologists to quickly adjust text while attending to PCPs' information needs. What we found, however, was that editing was not merely a matter of fixing words. Each edit became a site where different representations were negotiated: dermatologists' clinical reasoning, PCPs' expectations, and the LLM's latent understanding of medical communication.

The premise that we could completely offload the cognitive work of composing reports, and with it proportionally decrease the effort required, proved naive. Instead of eliminating work, the system redistributed it. Where reporting had previously meant composing text directly, it now required (1) entering structured diagnostic information, (2) supervising the LLM's generated output, and (3) performing coordination work across multiple representations. In distributed cognition terms [25, 28], the task of composing had shifted from the dermatologist's internal representation to an external system, but this shift did not reduce overall cognitive load. Rather, it introduced new coordination demands: clinicians now had to translate their clinical knowledge into structured input, anticipate how the model might interpret it, and scrutinise its output for errors, omissions, or overstatements. Even when hallucinations were rare, their possibility made supervision mandatory. For some participants, this made the system feel less like a tool relieving burden and more like an unnecessary addition without clear purpose, particularly when downstream benefits for PCPs remained more abstract than the immediate, tangible work required.

Participants repeatedly compared our prototype to their existing use of "smart phrases" or dot phrases, which are pre-defined macros that reliably expand into custom text. These offer consistency and efficiency at the cost of individuality, often producing reports that lack attentiveness to patient-specific details. Prior research shows that such templated notes can invoke a "copy-paste" feeling, leaving PCPs unconvinced that the dermatologist truly engaged with their case [70]. If minimising author effort were the only goal, macros and templates might already represent an optimal solution. Our prototype demanded more effort precisely because it pursued a different objective: extending the benefits of LLM assistance beyond the reporting dermatologist to include PCPs. By embedding PCPs' informational needs and perspectives into the reporting process, the system increased the number of representations requiring coordination. Once this broader ambition is accepted, coordination costs become unavoidable. Aligning across representations is the mechanism through which AI benefits extend beyond the primary user.

This reframing has important implications for designing LLM-based clinical tools. What seems like "wasted work" is often the glue that makes automation safe, usable, and trusted. Strauss described this as articulation work, i.e, the ongoing coordination needed to keep complex collaborative practices functioning [55]. Similarly, Elish and Watkins [19] called this type of work "repair work", which is needed to make AI-based systems contribute to broader clinical practice. Building on their thought, coordination should not be viewed solely as friction to minimise, but as productive work that enables better information flow across care pathways. The question shifts from "how can we reduce coordination effort?" to "how can we design coordination mechanisms that feel worthwhile?" This might mean making downstream benefits more visible to reporting clinicians, providing real-time feedback from PCPs about report quality, or offering coordination shortcuts for routine cases while preserving flexibility for complex ones. It also suggests that evaluating such systems purely on time-to-completion metrics misses the point. What matters is whether the coordinated output better serves the distributed work of patient care.

From this perspective, the effort participants spent coordinating representations was not wasted on personal preference or system limitations. It was invested in deliberate improvement of information continuity and support for care at the PCP level. HCI research suggests that users often value systems that support their work context, relational obligations, and interpretative boundaries as much or more than raw productivity [17, 29, 44, 49]. In this sense, using AI does not have to be about fitting more tasks into less time. By designing LLM-based automation that acknowledges and scaffolds this coordination work, rather than promising to eliminate it, we can build systems that strengthen communication across care pathways while respecting the complexity of collaborative clinical practice.

6.3 Addressing misalignments before the need for human-AI interactions

In this study, we deliberately introduced secondary stakeholder needs (the PCP template) into a clinical workflow where such needs had not previously been explicitly present. We anticipated that this could create disruptions, and indeed later conceptualised these as representational misalignments. Having established what representational design is and why it matters, we now consider how misalignments can be minimised before they require coordination. We see the following opportunities for designers, HCI researchers, and AI developers, which should be seen as complementary rather than isolated interventions.

Improving the fidelity of input clinical data. We found that our diagnostic form, structured to reflect the information required by the PCP template, was itself a source of misalignment. Although intentionally designed as a free-text form, participants sometimes resisted its predefined structure or found it misaligned with their habitual reporting practices. This tension echoes long-standing challenges in HCI and Health Informatics research on Electronic Health Records, where the introduction of structured documentation formats often disrupted clinicians' established workflows. Lessons from this literature can inform how to capture more comprehensive data without overburdening reporting clinicians. For example, Murray et al. proposed MedKnowts, a system that, among others, synthesises structured data from free-text clinical notes to support downstream computational use [38]. Similarly, Azadi et al. showed how structured data entry interfaces can be optimised for clinical reporting tasks [6]. More recent work on reporting explored how such input could be partially automated. Yildirim et al. demonstrated how multimodal AI models can generate keyword-based assessments from X-rays to scaffold reporting [64]. Such approaches could improve structured inputs for LLMs while preserving clinicians' preferred documentation styles. By improving the quality and structure of input data, whether through better forms, supportive interfaces, or AI-assisted entry, we can reduce the severity of representational misalignments before they occur.

Incorporating trusted sources into the generation pipeline. Several misalignments we observed stemmed from the latent representations of the LLMs and the resulting unpredictability of their outputs. This is a well-known challenge in the AI community, which has developed techniques to ground generation in trusted sources. One

such approach is Retrieval-Augmented Generation (RAG), which reduces hallucinations by conditioning model outputs on validated reference documents, such as approved clinical guidelines [42]. For example, Zhao et al. introduced MedRAG, a RAG-based system that improved diagnostic accuracy and reduced misdiagnosis rates compared to state-of-the-art models [73]. Building on this idea, custom libraries of vetted treatment plans or smartphrases could be incorporated as retrieval sources to further align outputs with clinical expectations. Importantly, this also raises the question of data provenance, long recognised as critical for establishing trust in clinical AI [61]. Within our own system, we began addressing this by visually linking keywords from the input form to specific parts of the generated text. Similar mechanisms could be extended to trace how external, trusted sources shape outputs. Strengthening provenance and grounding could thus reduce misalignments at their source before coordination becomes necessary.

Exploring specialised language models. The LLM we used often generated content beyond the scope of our task. While occasionally useful, it more often introduced risk through unnecessary elaboration or drift. For clinical reporting, what is needed is not broad general knowledge but reliable structuring of medical data. One promising direction is the use of Small Language Models (SLMs), which could be trained on smaller, curated corpora of trusted clinical data. Such models can be tailored to local contexts and institutional practices, reducing hallucinations and increasing reliability [20, 35]. Alternatively, targeted fine-tuning of existing models offers a complementary path to align outputs with domain-specific requirements without the overhead of training new models from scratch [36]. Both approaches could help reduce representational misalignments at their source by constraining the AI's latent space to clinically appropriate content.

Participatory prompt engineering. In our prototype's last iteration, we introduced a functionality to let the participants inspect and edit the system prompt, i.e., the LLM's primary directive. We hoped to minimise the misalignments before they occurred. In one case, a participant removed a brevity requirement, and the resulting reports aligned more closely with their internal representations. While ad-hoc, this highlights the potential for structured participatory approaches to reduce misalignments. Recent work offers methods to support such approaches [18, 45]. For example, Djeffal et al.'s Reflexive Prompt Engineering framework provides a responsible, iterative process for designing prompts [16], and PromptHive enables collaborative prompt authoring that connects domain knowledge with rapid iteration [43]. In clinical contexts, participatory prompt engineering could involve clinicians and PCPs to formalise phrasing templates and constraints, balancing local control with upstream representational designs. Done thoughtfully, this approach may reduce misalignments without requiring extensive model retraining.

These strategies illustrate a complementary set of approaches for reducing representational misalignments before coordination becomes necessary. By improving input fidelity, grounding generation in trusted sources, tailoring models to domain-specific contexts, and supporting participatory prompt engineering, HCI and AI researchers can proactively shape the AI output and its alignment with multiple stakeholders. This upstream focus may not replace the need for coordination since (1) even advanced LLMs exhibit the same underlying nondeterministic limitations [51, 57], and (2) representations tailored to secondary stakeholders may significantly diverge from those of the primary users. However, they can reduce the frequency and intensity of misalignments, ultimately leading to AI-based systems as elements of coherent healthcare pathways, enabling more reliable, trustworthy, and actionable clinical reporting.

7 Limitations

We acknowledge that differences between healthcare systems may influence clinicians' expectations and use of AI tools. Danish and Swedish participants operate within publicly funded systems with gatekeeping by PCPs, while US participants typically work in privatised settings with more direct specialist access [21, 58]. These differences were considered when analysing the feedback from the participants.

Moreover, this work has several limitations. First, the study's format may not have yielded the most informative results regarding the explored interaction techniques. Although introduced, participants may not have had sufficient time to adopt these techniques fully. Longer-term studies could provide deeper insights into their effectiveness. Second, while using distributed cognition helped us conceptualise representations, misalignments, and coordination, other theoretical lenses (both inductive and deductive) might offer different interpretations and highlight other facets of the interactions. Third, the free-text design of the diagnostic form, in a few instances, led participants to write almost complete reports rather than providing concise keywords. Exploring alternative input methods could better support data collection and workflow alignment, directing attention to automation and not typing. Fourth, this study examined how embedding the goals of secondary stakeholders into a system reshapes the work of primary users. We acknowledge that the next step would be to assess the quality of the produced reports with the secondary stakeholders, in our case, PCPs. Fifth, participatory prompt engineering was introduced late in the study. Starting with a co-designed approach could have aligned model outputs with participants' expectations from the outset, potentially revealing other insights. Finally, the ecological validity of our findings is limited by the study design, particularly the use of prefilled diagnostic assessments in a prototype setting. This allowed us to focus on report-generation coordination and the misalignments introduced by representational design. However, real-world teledermatology also involves diagnostic uncertainty, incomplete or ambiguous referral material, time pressure, and routine interruptions, all of which would likely intensify coordination demands. Our findings should therefore be read as a lower bound on the coordination work introduced by representational design rather than an upper bound. Future work could explore how the coordination demands identified in this study change in real-world settings.

8 Conclusions

This paper examines how designing LLM-based systems for secondary stakeholders reshapes primary users' work. Using a teledermatology reporting prototype, we studied how embedding downstream PCP information needs into the template structure influenced dermatologists' reporting practices. Our findings extend HCI

scholarship on multi-stakeholder AI and offer guidance for designers working in clinical contexts where AI outputs must serve multiple audiences.

Our central insight is the role of representational design in shaping automation. While prior work emphasises designing interfaces and workflows, we show that the representations embedded in AI systems (templates, prompts, and generation constraints) determine whose needs are prioritised and how primary users experience those choices. In our study, the PCP-derived template encoded downstream requirements but also generated additional coordination work for dermatologists as they negotiated among AI output, clinical judgement, and dual-audience expectations. This work surfaced across five alignment dimensions: factual accuracy, confidence calibration, clinical importance, completeness, and stylistic appropriateness.

We argue that this coordination is productive rather than problematic. To coordinate stakeholder perspectives while maintaining professional authority over clinical communication, dermatologists used four techniques: input revision, word-level editing, relevance pruning, and internal alignment. These practices ensured reports met both their own standards and PCP needs, illustrating how multi-stakeholder design produces interpretive work that specialists are well equipped to perform.

For HCI practitioners, our findings suggest three design directions. First, understanding misalignment dimensions helps anticipate where users will need coordination support: systems generating factually accurate but stylistically mismatched content require different interaction affordances than those producing incomplete but appropriately confident outputs. Second, representational design facilitates embedding stakeholder needs upstream, through templates, prompts, and generation constraints, complementing traditional interface-level design. Third, reducing coordination overhead requires attention to data quality (improving input fidelity), model grounding (anchoring outputs to trusted sources), architectural choices (considering domain-specific models), and participatory design (involving users in prompt engineering).

Overall, we show that multi-stakeholder AI design hinges on understanding how upstream representational choices create downstream coordination demands. By characterising these demands and the techniques users adopt to manage them, we provide a foundation for designing AI systems that extend benefits across stakeholder boundaries while respecting the professional judgement of those accountable for their outputs. While grounded in teledermatology, the alignment dimensions and coordination techniques identified here are likely to travel — across other clinical contexts such as radiology reporting and discharge summaries, and more broadly to any setting where LLM-generated outputs must simultaneously serve primary users and secondary stakeholders with different expertise, conventions, and information needs.

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